FPT UNIVERSITY

# Capstone Project Report

## Report 1 – Project Introduction

– Can Tho, October 2026 –

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# I. Record of Changes

| Date | A*<br>M, D | In charge | Change Description |
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# II. Project Introduction

## 1. Overview

### 1.1 Project Information

- Project name: Frevia – Verified Freelance Marketplace Platform
- Project code: SEP490_04
- Group name: Frevia
- Software type: Web Application

### 1.2 Project Team

| Full Name | Role | Email | Mobile |
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## 2. Product Background

The rapid growth of remote work has turned freelance collaboration into a default mode of software and digital service delivery. On large marketplaces, however, the same problems keep recurring. Clients cannot easily judge whether a candidate is who the profile claims to be, and they must commit money before any deliverable exists. Freelancers, on the other side, cannot easily verify that a client will pay, and they often work without a written scope, without a milestone schedule, and without a recoverable record of what was agreed. The result is a market where most of the effort is spent on risk mitigation rather than on the work itself, and where the tools that would mitigate that risk – identity verification, escrow, structured milestones, evidence-based dispute handling – are usually sold as separate products or reserved for enterprise customers.

FREVIA (Freelance Via) is a web-based freelance marketplace that addresses this problem by putting verification, escrow, and project governance into the core transaction rather than

into an optional add-on. The platform covers the complete engagement lifecycle: a client publishes a job with a fixed budget, deadline, required skills, and the number of positions to fill; freelancers submit proposals, either as drafts or as submitted applications with attachments and an expiry date; the client compares proposals and hires one candidate; a contract is generated from the accepted proposal, priced at the agreed total amount with a fixed platform fee recorded on the contract, and must be signed by both parties before work starts; the work itself is divided into milestones that are funded, submitted, reviewed, and released one at a time; and the engagement closes with a review that becomes part of both parties' public reputation.

Trust is enforced at several layers. Every freelancer can complete an electronic identity verification in which the front and back of an identity document, a selfie, and a short sequence of liveness frames are submitted to an AI verification service that performs optical character recognition, QR-code reading, image quality assessment, face matching, passive and active liveness detection, and tamper detection before returning a verified or rejected result. Verified documents are additionally reviewed by an administrator, so a freelancer profile can display an identity badge that is backed by evidence rather than by a self-declaration. Profile edits made by freelancers, including expert profiles, are submitted as revisions with a proposed data snapshot and a profile strength score, and only become public after administrative approval. On the financial side, client funds are held in escrow at the milestone level, so a freelancer can see that money exists before starting a milestone, and a client only releases funds after reviewing the submitted deliverable.

Disagreement is treated as a first-class process rather than an exception. When a milestone submission is rejected and the two parties cannot agree, either side can open a dispute on that milestone. The dispute carries an arbitration fee with its own payment deadline and a response deadline, both parties file structured evidence marked as claim or response, and an administrator issues a decision that splits the milestone amount between freelancer and client and records a written reasoning. A party that disagrees with the decision may request a review, after which a final decision is recorded and the dispute is closed. Every resulting money movement is written to a transaction ledger with a distinct type – escrow deposit, milestone payout, platform fee, dispute fee, arbitrator payout, or refund – so the financial history of a contract can be reconstructed.

Artificial intelligence is applied in three narrowly scoped places where it removes real manual work, and deliberately not in the place where it would take a decision away from the user. First, a freelancer can upload a PDF curriculum vitae and have an external AI service extract a list of skills together with an estimated proficiency level on a one-to-ten scale; the freelancer reviews the suggestion and decides what to add to the profile. Second, the identity verification described above is an AI pipeline rather than a manual document check. Third, forum content is screened by an AI moderation step that returns a score, matched categories, and a recommendation; every post still enters a pending state and becomes publicly readable

only after an administrator approves it. FREVIA does not implement an automated job-to-freelancer recommendation or matching engine, because hiring remains a human judgement that depends on context the platform does not observe, and because a ranking model trained on a young marketplace would encode the biases of a very small sample.

Collaboration during the engagement is supported by a real-time messaging layer built on WebSocket connections authenticated with the same access token as the rest of the API. A conversation is one-to-one between a client and a freelancer, supports text and file attachments, read receipts, typing indicators, per-participant hiding and pinning, and soft deletion of individual messages. Contracts additionally expose shared files, so that attachments do not have to live in a private chat history. A persistent notification record is written for every relevant domain event, and a community forum with categories, comments, likes, and reporting gives users a place for public discussion that is moderated rather than unmoderated.

The platform is bilingual. The interface is available in English and Vietnamese, selected through a locale cookie, and language switching does not change the URL structure, which keeps deep links stable when a user changes language. Access control is dynamic rather than hard-coded: a permission is a record that binds an HTTP method and a path pattern to a module, roles are granted permissions, and a single guard evaluates every request against the permissions of the role currently active in the session. This makes it possible for an administrator to adjust what a role can do without a redeployment, and it allows one account to hold several roles and switch between them, which matters for users who both hire and work.

Financially, FREVIA relies entirely on Stripe. Freelancers and clients connect a Stripe Express account through an onboarding link, and the onboarding status is tracked on the user record so the platform knows whether payouts are possible. Milestone funding can be started either as a PaymentIntent or as a hosted Checkout Session, and a saved card can be reused for later milestones. Released milestones are paid out as Stripe transfers, refunds are issued as Stripe refunds, and all asynchronous state changes arrive through a single webhook endpoint whose signature is verified against the raw request body. No blockchain component, wallet, or proprietary settlement infrastructure is involved; the transaction ledger is an application-level record that mirrors the state reported by Stripe.

FREVIA is therefore best understood as a governed marketplace rather than a listing board. Its value comes from the sequence it enforces – verified identity, signed contract, funded milestone, reviewed deliverable, recorded decision, rated outcome – and from the fact that every step of that sequence leaves a durable record that both parties and the platform administrator can inspect later.

## 3. Existing Systems

Several freelance platforms currently dominate the digital labour market. This section reviews the four platforms that are closest to FREVIA and then positions the proposed system against them. For each platform the review covers its scope, the actors it serves, its main features, its strengths, its limitations, and its general architecture family.

### 3.1 Upwork

- **Overview:** Upwork is a global freelance marketplace where clients publish jobs and freelancers submit proposals for hourly or fixed-price contracts.
- **Link:** https://www.upwork.com
- **Actors:** clients, freelancers, and platform administrators.
- **Main features:** job posting, submitted proposals, hourly and fixed-price contracts, milestones, messaging, work diaries, reviews, and payment protection.
- **Strengths:** a very large global user base, a mature reputation system, a verified payment method requirement, and a complete contract-to-payment workflow.
- **Limitations:** service fees are charged on both sides of the transaction, competition on popular postings is heavy, and talent discovery still depends mostly on manual proposal review.
- **Architecture:** web SaaS.

### 3.2 Fiverr

- **Overview:** Fiverr is a service catalogue built from predefined packages, called gigs, in which freelancers publish a fixed scope and buyers order directly.
- **Link:** https://www.fiverr.com
- **Actors:** buyers, sellers, and platform administrators.
- **Main features:** gig packages with fixed scope, orders, in-order messaging, delivery and revision cycles, and reviews.
- **Strengths:** extremely low friction for buyers, transparent pricing, and a catalogue model that shortens the path from search to purchase.
- **Limitations:** packaged scope fits poorly with exploratory or long-running projects; there is no bidding, no milestone schedule, and limited room for negotiated requirements.
- **Architecture:** web SaaS.

### 3.3 Freelancer

- **Overview:** Freelancer is a project marketplace with contests, bidding, and milestone-based payments across many categories.
- **Link:** https://www.freelancer.com
- **Actors:** clients, freelancers, contest holders, and platform administrators.
- **Main features:** contests, bidding, milestone-based payments, messaging, and reputation management across a broad set of categories.
- **Strengths:** broad category coverage, competitive bidding, and milestone escrow that is available to a worldwide user base.
- **Limitations:** clients must screen a high volume of low-quality bids, and dispute outcomes depend on a support process that is largely opaque to both parties.
- **Architecture:** web SaaS.

### 3.4 Toptal

- **Overview:** Toptal is a curated talent network that screens applicants and matches a small, vetted pool of freelancers with clients through account managers.
- **Link:** https://www.toptal.com
- **Actors:** clients, vetted freelancers, account managers, and platform administrators.
- **Main features:** applicant screening, talent matching, account management, and contract and payment handling.
- **Strengths:** a high signal-to-noise ratio for clients, strong screening, and a professional, low-risk engagement model.
- **Limitations:** selective admission excludes most freelancers, pricing is high, and the matching decision is largely made by the platform rather than by the client.
- **Architecture:** web SaaS.

### 3.5 FREVIA (Proposed System)

- **Overview:** FREVIA is a web-based freelance marketplace that places verification, escrow, and project governance at the centre of the transaction rather than in an optional add-on.
- **Actors:** clients, freelancers, experts, and platform administrators.

- **Main features:** email and Google authentication with one-time-code verification, multi-role accounts with a dynamic permission model, freelancer, client, and expert profiles with administrative revision approval, AI-assisted identity verification, job posting with fixed-price budgets and multiple hiring positions, proposals with drafts and expiry, dual-signed contracts with a fixed per-contract platform fee, milestone funding and release through Stripe, escrow-backed dispute arbitration, contract reviews with responses, one-to-one real-time messaging, in-app notifications, an AI-moderated community forum, and advertisement banners.
- **Strengths:** verification and escrow are part of the standard transaction instead of an upgrade: identity is checked by an AI document and liveness pipeline, money is funded per milestone before work begins, and disagreement follows a documented arbitration process with evidence, deadlines, a split decision, and a review step. The permission model is data-driven and can be changed without redeployment.
- **Limitations:** the platform is new and has no established user base; payouts require Stripe Connect availability in the participant's country; only fixed-price contracts in US dollars are supported; and artificial intelligence is limited to CV skill extraction, identity verification, and forum moderation, with no automated recommendation or matching engine.
- **Architecture:** web application with external AI services.

The review shows where FREVIA differentiates itself and where it deliberately does not compete. Upwork and Freelancer already provide milestone escrow, and Toptal already provides strong vetting, so FREVIA does not claim to invent either concept. What the existing platforms generally do not provide is a transparent arbitration procedure that both parties can drive themselves, an identity check whose evidence is retained and reviewable by an administrator, and a permission model that lets a marketplace operator reshape role capabilities at runtime. Those three properties, together with the decision to keep hiring under human control, define the gap that FREVIA targets.

## 4. Business Opportunity

The freelance market has grown faster than the infrastructure that makes it safe. Most platforms solved the discovery problem years ago: a client can publish a job in minutes and receive proposals within days. What remains unsolved is the settlement problem. A client still has to transfer money to someone whose identity is asserted rather than demonstrated, and a freelancer still has to start work on the promise of a future payment. The cost of that uncertainty is paid in every transaction, through higher prices for clients, delayed payment for freelancers, and a large volume of work that never leaves the platform's support queue because there is no shared record of what was agreed.

This creates three concrete openings that FREVIA is designed to occupy.

The first is verifiable identity. A marketplace that keeps evidence of an identity check, together with the result of an AI document and liveness pipeline and an administrator's review, can let a freelancer carry a badge that means something. This is especially relevant for cross-border engagements, where the two parties share no legal jurisdiction, no mutual contact, and no practical way to verify each other before work begins. FREVIA stores the verification outcome and its supporting documents, exposes only the resulting status to other users, and keeps the documents accessible to administrators and to the document's owner. The same mechanism can later support compliance obligations that a marketplace operator may be required to demonstrate.

The second is escrow at a granular level. Clients are far more willing to commit money when the commitment is bounded. FREVIA asks the client to fund one milestone at a time rather than the whole contract, and it publishes the resulting payment status – pending, funded, released, refunded, or disputed – on the milestone itself. A freelancer can therefore see that the money for the current milestone has already been collected before starting work, while the client retains the ability to review the deliverable and to request changes with a stated due date before releasing funds. This structure converts a single high-stakes payment decision into a series of low-stakes ones, which is precisely the change that makes a first engagement between strangers possible.

The third is a dispute procedure that both parties can operate. On most platforms a dispute disappears into customer support, and neither party can predict the outcome, the timeline, or the evidence that will be considered. FREVIA makes the procedure explicit and symmetric: the dispute is attached to a milestone, both parties owe a response within a stated deadline, both may upload evidence tagged as a claim or a response, an arbitration fee is charged with its own payment deadline, and the administrator's decision records a split of the milestone amount together with a written reason. The losing party can request a review, which produces a final decision. Because the whole sequence is stored, the outcome is defensible after the fact, and the threat of a well-documented dispute discourages the behaviour that makes disputes common in the first place.

Reliability is not only a matter of money. A large share of collaboration failure comes from communication that is scattered across private channels. FREVIA keeps the conversation inside the contract: one-to-one real-time messaging with attachments, read receipts and typing indicators, and shared files attached to the contract rather than to a chat thread. A persistent notification record ensures that a proposal decision, a milestone submission, a payment event, or a dispute deadline is not lost because nobody opened a particular page. A community forum, with an AI moderation step and mandatory administrative approval before publication, adds a public layer where users can ask questions and compare experiences.

The AI strategy is deliberately narrow, and that narrowness is itself a commercial position. FREVIA uses AI where the output is verifiable and reversible: extracting candidate skills from an uploaded CV so that a freelancer does not have to type them, verifying an identity document and a live face so that an administrator does not have to inspect every upload, and scoring forum content so that a moderator's queue is ordered by risk. It does not use AI to decide who should be hired. That decision is left to the client, with the platform supplying structured, comparable information – skills with proficiency levels, portfolio items, verified identity status, completed contracts, and ratings – instead of an opaque score. For a marketplace that is still building trust with its first users, an explainable process is worth more than an automated one.

The bilingual requirement reinforces the same positioning. FREVIA ships in English and Vietnamese, and the language is a user preference stored in a cookie rather than part of the URL, so switching language does not break a shared link to a job or a contract. For a platform launched from Vietnam and aimed at freelancers who work for international clients, this is not a localisation afterthought but a primary adoption path: the same users need to read job descriptions in English and to operate the contract, milestone, and dispute screens in Vietnamese.

Taken together, these choices define a market position that is narrower than Upwork's and deeper than a listing board's. FREVIA targets engagements where both parties need a defensible record: a verified identity on one side, funded escrow on the other, and a written procedure for the case where the two disagree. That segment is small enough to serve with a focused product and large enough to validate the model before any expansion into hourly contracts, multi-currency settlement, or additional verification methods.

## 5. Software Product Vision

FREVIA is a web-based freelance marketplace built around a single proposition: an engagement between two strangers should be governable from beginning to end without either party having to trust the other on faith. The product therefore does not compete on the size of its talent pool. It competes on the completeness of the record it produces for each engagement – a verified identity, a signed contract, a funded milestone, a reviewed deliverable, and, when the parties disagree, a documented arbitration decision.

The primary users are clients who need a bounded, verifiable engagement and freelancers who need to know that payment exists before work begins. A third role, the expert, provides a public professional profile with an experience and certification record, intended for users whose value lies in review and consultation rather than in delivery. A fourth role, the administrator, is an operator rather than a customer: it approves profile revisions, reviews identity documents, orders the moderation queue, resolves disputes, and adjusts what each role is

permitted to do. The same person may hold more than one role, and the interface follows the role that is currently active in the session.

For the client, the vision is a hiring flow that never requires a leap of faith. Posting a job captures a fixed budget range, a deadline, the required skills, the categories, and the number of positions to fill. Proposals arrive with the freelancer's profile, skill proficiency, and attachments, and can be compared side by side before a candidate is hired. Hiring generates a contract that both parties must sign; funding a milestone moves the amount into escrow; approving a submission completes the milestone and makes the funded amount releasable, after which the client releases it as a payout and settles the fixed platform fee. At any point the client can see the money state of the contract, the status of each milestone, and the files that have been exchanged.

For the freelancer, the vision is the mirror image. A profile is built from structured evidence: a title, education, certifications, languages, work experience, skills with a proficiency level, portfolio items with technologies and a project link, and a curriculum vitae that can be parsed to propose skills automatically. Identity verification turns a claim into a badge. Job discovery is driven by explicit filters and saved searches rather than by an opaque ranking, bookmarks keep interesting postings within reach, and proposals can be prepared as drafts, submitted with attachments and an expiry date, edited, or withdrawn. Once hired, the freelancer can see which milestones are funded, submit work with files and links, respond to a change request with a due date, and escalate to a dispute if the review does not converge.

The organising principle behind these two journeys is that every state transition must be visible to both sides and must be reversible only through a recorded action. A milestone is not silently completed; it is submitted, optionally returned for changes, and then approved. A contract is not implicitly accepted; both signatures are stored. A dispute is not resolved by an email; the split, the reasoning, and the reviewing administrator are all persisted. This is why the platform invests in a transaction ledger, a notification record, and an explicit status model rather than in a chat-first experience.

Artificial intelligence supports this principle rather than replacing it. The AI components in FREVIA are narrow by design: a CV analyser that proposes skills with a proficiency estimate, an identity verification pipeline that reads a document, matches a face, and checks liveness, and a forum moderation step that scores content before a human approves it. None of them makes a decision that binds a user. The client still chooses whom to hire, the administrator still approves or rejects, and the freelancer still confirms what goes onto the profile. Keeping a human at each decision point is what allows the platform to explain any outcome to the person affected by it.

The technical vision follows the same logic. FREVIA is a monorepo that holds a NestJS API, a Next.js web client, a shared package of validation schemas and generated types, and

a shared user-interface package. Because the validation schemas are shared, a rule such as a minimum proposal length or a maximum milestone amount is enforced identically in the browser and on the server. Access control is data-driven: permissions are rows that bind an HTTP method and a path to a module, roles are granted permissions, and a single guard checks every request against the role active in the session, which means authorisation can be corrected in production data rather than in code. Money is handled exclusively through Stripe – Express onboarding for payouts, PaymentIntents and hosted Checkout Sessions for funding, transfers for release, refunds for reversal, and a signature-verified webhook for asynchronous state. The system is containerised so that a fresh environment can be started with one command.

The longer-term direction is to deepen rather than to widen. The most valuable extensions are the ones that reuse the existing record: additional verification methods behind the same document and liveness interface, hourly contracts alongside the current fixed-price model, multi-currency settlement once Stripe support allows it, and richer reputation signals derived from completed contracts rather than from self-reported claims. Each of these extends a structure that already exists, which is the practical test the team applies when deciding what belongs on the roadmap.

## 6. Project Scope & Limitations

### 6.1 Major Features

The feature list below describes what the delivered system actually exposes, grouped by functional area. Each identifier is referenced by the estimation table in the Project Management Plan.

***Identity and Access***

**FE-01** Email and password registration with role selection and one-time-code email verification.

**FE-02** Google OAuth sign-in, and linking a Google identity to an existing account.

**FE-03** JWT access and refresh token authentication backed by persisted session records containing device information, IP address, and expiry.

**FE-04** Session management: listing active sessions, viewing session details, and revoking a session.

**FE-05** Password reset through an emailed one-time code, and password change from account settings.

**FE-06** Multi-role membership: one primary role, joining an additional role, and switching the active role within a session.

**FE-07** Data-driven role-based access control in which a permission binds an HTTP method and path pattern to a module, and roles are granted permissions.

**FE-08** Administrative role and permission management, including role creation and permission assignment or removal.

**FE-09** Administrative user management: creating users, updating account details, and banning or unbanning accounts.

***Profiles and Portfolio***

**FE-10** Account profile with avatar and cover upload, display name, biography, online status, availability status, and profile completion percentage.

**FE-11** Freelancer profile with professional title, education, certifications, languages, and work experience entries.

**FE-12** Skills carrying a proficiency level from one to ten, with skill suggestions and add or remove operations.

**FE-13** Portfolio items with description, technologies, external project link, and cover image.

**FE-14** Curriculum vitae upload, preview, replacement, and deletion for PDF files up to ten megabytes.

**FE-15** AI-assisted curriculum vitae analysis that proposes skills with estimated proficiency levels for the freelancer to review before adding them.

**FE-16** Profile strength and completion scoring used to guide profile improvement.

**FE-17** Profile revision workflow in which edits are stored as a proposed snapshot and become public only after administrative approval.

**FE-18** Client profile with company name, company description, and website.

**FE-19** Expert profile with expertise areas, years of experience, education, certifications, and website, together with a public expert directory.

**FE-20** Social links, favourite freelancers, follow and unfollow, freelancer discovery, and public profile pages.

*Identity Verification*

**FE-21** Identity verification submission carrying identity document front and back images, a selfie, and a sequence of liveness frames.

**FE-22** AI identity verification covering optical character recognition, QR code reading, image quality assessment, face matching, passive and active liveness detection, and tamper detection.

**FE-23** Administrative review of identity verification documents, and display of a verified identity badge on the profile.

*Job Marketplace*

**FE-24** Job creation with description, fixed-price budget range, deadline, required skills, categories, hiring type, and number of positions.

**FE-25** Job editing, deletion, and status lifecycle across draft, open, in progress, completed, closed, and cancelled.

**FE-26** Stable slug-based public job URLs that remain valid when a posting is edited.

**FE-27** Public job search with keyword matching and multi-criteria filtering by category, skill, budget range, and creation date, with configurable sorting.

**FE-28** Job detail page exposing client information, required skills, and contextual actions for the current user.

**FE-29** Job bookmarks with save, remove, list, and status lookup.

**FE-30** Saved searches that persist a filter configuration for later reuse.

**FE-31** Client job dashboard with status and search filters, and applicant counts per posting.

**FE-32** Public browsing of job categories.

*Proposals and Contracts*

**FE-33** Proposal drafting and submission with bid amount, estimated duration, cover letter, and file attachments.

**FE-34** Proposal editing, withdrawal, and expiry handling before a hire decision.

**FE-35** Freelancer proposal tracking with status filters.

**FE-36** Client proposal review for a job, including comparison of applicant profiles and attachments.

**FE-37** Hiring or rejecting a proposal, where hiring generates a contract linked to that proposal.

**FE-38** Contract list and detail for both client and freelancer.

**FE-39** Dual contract signing, recording the client and freelancer signatures separately.

**FE-40** Contract total amount and a per-contract platform fee, recorded with its own payment status and set to USD 10.00 by default.

**FE-41** Contract completion and cancellation.

**FE-42** Milestone creation, editing, deletion, and ordered listing within a contract.

**FE-43** Milestone payment state tracking across pending, funded, released, refunded, and disputed.

**FE-44** Milestone submission with a message, external links, and uploaded files.

**FE-45** Change requests carrying a message and a due date, and submission approval that completes the milestone and makes its funded amount releasable.

**FE-46** Milestone file management and contract-level shared files with upload, download, and deletion.

**FE-47** Milestone start transition from pending to in progress, restricted to the assigned freelancer and limited to one active milestone per contract.

***Payments***

**FE-48** Stripe Connect Express onboarding through an account link, with onboarding status tracked on the user record.

**FE-49** Saved payment methods and a Stripe customer portal session for managing cards.

**FE-50** Platform fee payment initiated as a PaymentIntent, a hosted Checkout Session, or a saved card charge.

**FE-51** Milestone funding through a PaymentIntent, a hosted Checkout Session, or a saved card.

**FE-52** Milestone release as a Stripe transfer to the freelancer account.

**FE-53** Refund processing for a funded milestone.

**FE-54** Dispute fee payment with its own deadline, and dispute settlement with split payouts.

**FE-55** Transaction ledger recording type, status, amount, and Stripe references for every money movement.

**FE-56** Manual synchronisation of PaymentIntent and Checkout Session state with the local record.

**FE-57** Stripe webhook endpoint with raw body signature verification.

***Disputes and Reviews***

**FE-58** Dispute creation on a milestone, with claim evidence attached.

**FE-59** Dispute response with evidence uploaded before the response deadline.

**FE-60** Dispute list and detail for the two participants, including fee and deadline status.

**FE-61** Decision review request as an escalation step after an administrative decision.

**FE-62** Administrative arbitration producing a split of the milestone amount, a written reasoning, and a final decision.

**FE-63** Contract review with an overall rating and a category breakdown.

**FE-64** Review editing, deletion, and contract-level review listing.

**FE-65** Review responses and management of those responses.

***Communication and Notifications***

**FE-66** Conversation list with unread counts, pinning, and hiding on a per-participant basis.

**FE-67** Starting a conversation between a client and a freelancer.

**FE-68** Real-time messaging over an authenticated WebSocket connection, with text messages, read receipts, typing indicators, and message deletion.

**FE-69** File attachments in messages with name, size, and type metadata.

**FE-70** Notification list with typed entries and an unread badge.

**FE-71** Marking notifications as read individually or all at once, and deleting notifications.

*Community Forum*

**FE-72** Forum browsing across categories, with search, category filtering, and sorting.

**FE-73** Forum post creation with a rich text editor and tags, plus editing and deletion by the author.

**FE-74** Comments, likes on posts and comments, and reporting of posts and comments.

**FE-75** AI moderation scoring attached to a forum post, with every post held in a pending state until an administrator approves it.

**FE-76** Administrative moderation queue, report handling, and trash management for posts and comments.

*Administration and Platform*

**FE-77** Administrative management of job categories, including status changes, soft deletion, and restore.

**FE-78** Administrative management of the skill catalogue, including soft deletion and restore.

**FE-79** Advertisement banner management across five placement positions, and display of active banners on the public site.

**FE-80** Administrative dashboards for identity verification review and profile revision review.

**FE-81** Bilingual interface in English and Vietnamese with cookie-based locale selection that leaves URLs unchanged.

**FE-82** Light and dark themes with a responsive layout across desktop and mobile widths.

**FE-83** Shared request and response validation schemas between the web client and the API, a standard response envelope, a global exception filter, and containerised deployment of both applications with health checks.

### 6.2 Limitations & Exclusions

**LI-01** All payment and payout functionality is provided by Stripe. Direct bank integration, domestic e-wallets, mobile money, and blockchain or cryptocurrency settlement are outside the scope of the current implementation.

**LI-02** Stripe Connect onboarding and payouts depend on the availability of supported Stripe account countries and capabilities. The platform does not hold balances itself, does not operate an internal settlement ledger as a system of record, and cannot pay out to a country that Stripe does not support.

**LI-03** Only fixed-price contracts are supported. Hourly contracts, time tracking, work diaries, and automatic billing based on recorded hours are excluded from the current scope.

**LI-04** Monetary amounts are handled in United States dollars only. Currency conversion, multi-currency contracts, and locale-specific tax calculation are excluded.

**LI-05** Artificial intelligence is limited to three functions: curriculum vitae skill extraction, identity document and liveness verification, and forum content moderation. No recommendation, ranking, or automatic matching engine is implemented, and no AI component makes a hiring, approval, or arbitration decision on its own.

**LI-06** The AI functions depend on two external services reachable over HTTP. When either service is unavailable, the affected action fails with an explicit error and administrative manual review remains the fallback; response quality for documents that are poorly photographed, rotated, or outside the supported document types is not guaranteed.

**LI-07** Identity verification is tuned to Vietnamese identity documents. Passports, driver licences, and identity documents from other jurisdictions may not be recognised reliably, and no alternative verification channel is provided.

**LI-08** Arbitration is an administrative decision made by a platform administrator, not a legal proceeding. The system does not enforce jurisdiction-specific law, does not provide binding legal arbitration, does not issue enforceable awards, and does not handle debt collection or chargeback litigation.

**LI-09** The product is delivered as a responsive web application. Native iOS and Android applications are not part of the current scope.

**LI-10** One-time codes and account notifications are delivered by email through an SMTP provider. SMS delivery, authenticator application support, and push notifications are excluded; if email delivery is delayed or blocked, account verification and password reset are affected.

**LI-11** Several capabilities exist in the data model but are intentionally not exposed through the current release: time-based one-time password two-factor authentication, invoice documents, completion certificates, reusable contract term templates,

job alerts, recently viewed jobs, project progress updates, portfolio likes, system settings management, a banned keyword list, and announcement banners.

**LI-12** Notification records are persisted and retrieved by the client, but notification events are not pushed over a real-time channel. New notifications become visible on the next fetch or page load.

**LI-13** The deployment uses a single API instance with in-memory WebSocket room state. Automated continuous integration, horizontal scaling behind a load balancer, sticky sessions, and a shared adapter for the real-time layer are not implemented, so the current topology does not support running several API replicas concurrently.